

Alexander F. Vasquez

Lima, Perú

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Research Interests

My research interests lie at the intersection of model-based and learning-based control, with a focus on how optimization-based methods can be integrated with learned models and perception pipelines to develop safe and capable controllers for autonomous robotic systems.

Education

Pontificia Universidad Católica del Perú

July 2021 – Present

BS in Mechatronics Engineering

- **Overall Grade:** 17.6/20
- **Relevant Coursework:** Dynamic Systems and Control I & II, Robotics, Artificial Intelligence, Nonlinear Control, Optimal Control, System Identification.

Honors

Beca 18 Scholarship

June 2021

PRONABEC

Awarded a full scholarship for undergraduate studies at the Pontificia Universidad Católica del Perú, granted to low-income students with outstanding academic performance.

**Recognition for Best Performance and Top Positions in EEGGCC –
2023-1 & 2023-2**

August 2023

June 2024

Pontificia Universidad Católica del Perú

Second Place in General Science Studies during the 2023-1 and 2023-2 semesters

Research Experience

Summer Undergraduate Research Fellow (SURF)

Pasadena, CA

California Institute of Technology (Caltech) – AMBER Lab

Jun 2026 – Aug 2026

- Designing and integrating a 2-DOF active vision module (pan-tilt camera head) into the Unitree G1 humanoid robot at Caltech's AMBER Lab, extending the robot's observable workspace through dynamic gaze control for visual tracking during locomotion.
- Developing a control stack combining MPC-CBF for safe navigation, RL-based locomotion policies (mjlal/RSL-RL), and a YOLO11n + ByteTrack perception pipeline for real-time target tracking, validated first in MuJoCo simulation ahead of hardware integration with CubeMars AK40-10 actuators.
- Working on reinforcement learning and imitation learning for humanoid locomotion and sports-related skills, including ball-kicking behaviors in simulation.
- Tools: ROS2, YOLO11n, ByteTrack, MuJoCo/mjlal, RSL-RL, CasADi, OSQP, Python, PyTorch.

2-DOF Bicopter for Thesis

[\[video\]](#) 

- Design and implementation of a bicopter module for Dynamic Systems and Control I & II courses, proposed for future use in advanced control courses.
- Tools Used: Inventor, Bambu Studio, C++, Eagle, MATLAB, Simulink

Control System for 2-DOF Bicopter

[\[Paper Draft and Code\]](#) 

- Developed in the course *Advanced Control for Mechatronic Engineering*; includes the design of Backstepping, Gain Scheduling, Feedback Linearization, and EKF Controllers.
- Tools Used: MATLAB, Simulink, ODE45

Publications

Design of an Agricultural Robot for Indoor Quinoa Seeding Using Visible Light Positioning

Apr 2026

Daniel Tomas Menacho Ordoñez, **Alexander Vasquez**, Rodrigo Carbajal, Jasper-Jan Lut, Jose Balbuena, Marco Zúñiga Zamalloa, Diego Quiroz

In: Advances in Automation and Robotics Research (LACAR 2025), Springer, pp. 345–354, 2026

DOI: [10.1007/978-3-032-18982-0_32](https://doi.org/10.1007/978-3-032-18982-0_32) 

Projects

Modular Robotic Platforms

[\[video\]](#) 

- Development of a 4-DOF robotic arm and an omnidirectional car for the Robotics and Artificial Intelligence course laboratory.
- Tools Used: C++, Inventor, 3D Printing

Educational Platforms for Control and Automation

- Implementation of modules including a robotic arm, pick-and-place systems, and quality inspection processes.
- Tools Used: C++, Ladder Logic, MATLAB, Simulink, S7-PLCSIM

Balancing Robot for Dynamic Systems and Control I & II

[\[video\]](#) 

- Design and implementation of a self-balancing robot.
- Tools Used: Inventor, 3D Printing, Thonny, MicroPython, Linux, MATLAB, Simulink

Design of an Agricultural Robot for Indoor Quinoa Seeding

- Collaborated on the development of a robotic platform for indoor quinoa seeding using visible light positioning.
- Contributed to the implementation of the color grid used for localization and participated in the prototype assembly and testing to validate the navigation system.
- Tools Used: Python, MATLAB, Arduino IDE, Inventor, 3D Printing

Design and Implementation of a Hybrid Ground–Aerial Robot

[\[video\]](#) 

- Developed in the course *Mechatronic Design Project*; design and implementation of a robot capable of transforming from a ground vehicle to an aerial platform.
- Tools Used: Pixhawk Firmware, Arduino IDE, Teensy 4.1, C++, Inventor, Eagle, 3D Printing

Teaching and Professional Experience

Teaching Assistant in Digital Design

Lima, Peru

Pontificia Universidad Católica del Perú

Aug 2024 – Dec 2025

- Assisted students during laboratory sessions, supporting them in the implementation and debugging of code related to the course content.

Engineering Development Assistant

Lima, Peru

L.A. Bear Robotics & Automation S.A.C.S.

Jan 2025 – Jul 2025

Assisted in the design and development of engineering modules focused on practical training in control and automation areas. Key activities include:

- Development of control modules, such as water level control in tanks and the inverted pendulum.
- Integration of Simulink and microcontrollers for the control of the modules.
- Assembly of automation modules, including a robotic arm, pick-and-place systems, and quality inspection processes.
- Integration of subsystems and validation of their performance to ensure functionality and reliability of the modules in educational environments.

Teaching Assistant in Robotics*Pontificia Universidad Católica del Perú**Lima, Peru**Mar 2025 – Jul 2025*

- Assisted students during laboratory sessions, supporting them in the assembly of an omnidirectional robot, as well as in the implementation and debugging of autonomous navigation algorithms.

Teaching Assistant in Optimization and Control*Pontificia Universidad Católica del Perú**Lima, Peru**Jan 2026 – Mar 2026*

- Developed laboratory code and evaluation materials for LP, QP, LQR, MPC, and RCAC.
- Implemented numerical examples in MATLAB and assisted students during laboratory sessions.

Teaching Assistant in Dynamic Systems and Control*Pontificia Universidad Católica del Perú**Lima, Peru**Mar 2026 – Jul 2026*

- Developed laboratory code and evaluation materials for PID controllers, state observers, LQR, and LQG.
- Implemented numerical examples in MATLAB and Simulink, and assisted students during laboratory sessions for two parallel course sections (A and B).

Technical Skills

Programming: MATLAB, Simulink, C++, C, Python, MicroPython, VHDL, Verilog**Robotics Frameworks:** ROS2, MuJoCo, Pinocchio**Machine Learning:** PyTorch, Ultralytics YOLO, FastSAM, ByteTrack**Optimization & Control:** CasADi, IPOPT, OSQP, Eigen**Software:** Autodesk Inventor, Eagle, Siemens PLCSIM, Visual Studio Code, Thonny**Embedded Platforms:** NVIDIA Jetson Orin Nano, Arduino, STM32, Raspberry Pi, FPGA DE10-Lite**Languages:** Spanish: Fluent (Native), English: B2 (FCE)